

Two is Better than One: Efficient Ensemble Defense for Robust and Compact Models

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Abstract

Deep learning-based computer vision systems adopt complex and large architectures to improve performance, yet they face challenges in deployment on resource-constrained mobile and edge devices. To address this issue, model compression techniques such as pruning, quantization, and matrix factorization have been proposed; however, these compressed models are often highly vulnerable to adversarial attacks. We introduce the **Efficient Ensemble Defense (EED)** technique, which diversifies the compression of a single base model based on different pruning importance scores and enhances ensemble diversity to achieve high adversarial robustness and resource efficiency. EED dynamically determines the number of necessary sub-models during the inference stage, minimizing unnecessary computations while maintaining high robustness. On the CIFAR-10 and SVHN datasets, EED demonstrated state-of-the-art robustness performance compared to existing adversarial pruning techniques, along with an inference speed improvement of up to 1.86 times. This proves that EED is a powerful defense solution in resource-constrained environments.

1. Introduction

Convolutional Neural Networks (CNNs) have shown remarkable performance across various computer vision tasks, such as image classification, object detection, and semantic segmentation[18, 33, 36]. However, CNNs have become increasingly wider and deeper to achieve higher performance, leading to complex architectures with vast numbers of parameters. This complexity inherently results in significant computational costs and storage demands, making deployment on resource-constrained edge devices challenging[4, 42]. To address this challenge, a number of model compression techniques, including pruning[14, 24], quantization[3], and matrix decomposition[8], have been proposed to minimize model size and computational load

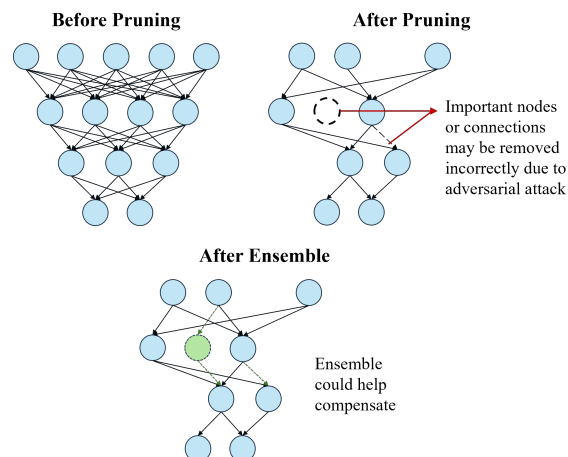


Figure 1. Challenges in AP and the potential of ensemble methods: AP weakens the network by removing critical parameters, reducing robustness and increasing vulnerability to adversarial attacks. An ensemble approach mitigates pruning effects by combining pruned models, enhancing robustness through their collective strength (illustrated in green).

without compromising accuracy.

Meanwhile, deep learning models are highly vulnerable to adversarial attacks that induce false predictions by disturbing the input image with small perturbations[11, 37]. This vulnerability is particularly pronounced in sparsely compressed models, which are even more sensitive to adversarial attacks than standard models[35, 44]. Consequently, common adversarial defense techniques such as adversarial training (AT) [20, 27], label smoothing [41, 51], and defensive distillation[31], while generally effective, often experience a degradation in performance when applied to compressed models. This indicates that existing compression techniques may compromise adversarial robustness even if they maintain accuracy on clean test sets. In response, adversarial pruning (AP), a model compression technique that considers adversarial robustness[34, 48, 52], has emerged;

however, AP still exhibits lower robustness compared to non-compressed defense models[32].

We assume that this degradation in robustness arises from reduced generalization capability and information loss caused by pruning. Specifically, adversarial attacks may cause the model to misidentify and even remove critical nodes or connections during the pruning process. In Sec.3.3, we validate this assumption by observing importance scores, which reflect how each component contributes to the loss function through the gradients of the weights[50]. Our findings demonstrate that adversarial examples significantly impact the importance scores of individual parameters.

Ensemble defense[7, 39] has recently gained attention in adversarial defense research for enhancing the stability and robustness of multiple base classifiers against attacks. As illustrated in Fig.1, we believe that ensemble defense methods can help mitigate generalization degradation and information loss during the pruning process. Since ensemble learning can compensate for deficiencies in individual compressed models due to removed parameters by other models and offset individual errors by combining outputs. However, traditional ensemble defenses inherently suffer from a large capacity, as they require multiple base models[15, 30]. This raises an important question: *Is it possible to develop an ensemble defense algorithm that not only achieves robustness but also results in a compact compressed model?*

As an answer to this, we propose Efficient Ensemble Defense (EED). Instead of using multiple base models, we compress a single base model by employing different pruning metrics and then utilize these compressed sub-models in the ensemble. This approach reduces the overall capacity compared to traditional ensemble defense models that rely on multiple base models. Given that using smaller models in an ensemble can lead to reduced defensive efficacy[46], we enhance the variety of the compressed models by incorporating a diversity term in the ensemble selection process, thereby improving ensemble effectiveness while maintaining robustness. Additionally, by adopting inference-efficient ensemble[25], we dynamically add sub-networks and perform ensemble operations as needed during inference, improving overall efficiency.

The main contributions of our work are as follows:

- Propose EED, that overcomes the limitations of traditional AP by enhancing the ensemble diversity and compensating for individual errors, thereby achieving robustness against adversarial attacks while being compact.
- Reduce memory by using a single base model in ensemble, and decrease computational costs by dynamically adding models only when necessary during inference.
- Achieve state-of-the-art performance across various attack benchmarks on the CIFAR-10 and SVHN datasets.

2. Related Works

2.1. Adversarial Attack and Defense

Adversarial attacks exploit the inherent vulnerabilities of deep learning models by adding specific noise to the input, thereby inducing incorrect predictions. These specially crafted inputs, designed to deceive deep learning models, are called adversarial examples. Early studies highlighted the susceptibility of deep learning models to adversarial examples[37]. Then, more potent attacks such as FGSM[11], PGD[27], and AutoAttack[6] have emerged.

On the other hand, defense methods against attacks have also been evolved. The most well-known adversarial training (AT)[1, 20, 27] enhanced robustness by training adversarial examples. Other strategies involve regularizing geometric properties such as the curvature of the loss function [41, 45], or restoring potentially attacked images to purified versions[29, 49]. Adversarial pruning (AP) and ensemble defense are also one of adversarial defenses (details are given in Sec.3.1.)

2.2. Adversarial Pruning

AP is a pruning technique (primarily weight pruning) that considers the adversarial context, aiming to reduce the complexity of neural networks while preserving robustness against adversarial attacks. HYDRA demonstrated that empirical adversarial robustness can be achieved through a lowest weight magnitude (LWM) pruning method. R-ADMM[48] and HARP[52] incorporated the alternating direction method of multipliers (ADMM) framework into LWM-based pruning to enhance robustness. In some studies [12, 22, 26], a traditional pruning method, i.e., the Lottery Ticket Hypothesis, was integrated with AT. However, they all failed to maintain generalized robustness at higher sparsity rates (s_r)[13, 19, 38], and struggled to defend against adaptive attacks[9, 16]. This suggests that conventional AP has not fully resolved the trade-off between model capacity and robustness.

2.3. Ensemble Defense

Ensemble defense (ED), which applies ensemble learning to adversarial defense, has been actively studied due to its promising performance[7, 39]. The core of ED is to improve predictions through an ensemble diversity loss that considers adversarial robustness during training. ADP[30] used Shannon entropy and geometric diversity for uncertainty, while GAL[17] utilized cosine similarities between model loss gradients to formalize diversity. DVERGE[47] introduced vulnerability diversity to ensure that each model is robust to the weaknesses of others, and MOL[5] increased perturbation diversity by using expert models trained on different attacks. Thus, ED has significantly enhanced robustness by reducing the risk of overfitting to adversarial ex-

amples and diversifying the decision boundary, though with increased model capacity and computational costs.

3. Preliminaries

3.1. Notations

Adversarial Attack Let $x \in \mathbb{R}^d$ be an input example, $y \in \{1, \dots, C\}$ the true label, and $f(\cdot)$ a classifier with parameters θ . The classifier's prediction for input x is given by $f(x) = \arg \max_{i \in \{1, \dots, C\}} f_i(x)$, where $f_i(x)$ denotes the predicted probability for class i . An adversarial example x' is crafted by adding a perturbation δ to x , aiming to maximize the model's prediction error while keeping x' close to x under a norm constraint. Formally, an adversarial attack is defined by:

$$x' = x + \delta = \arg \max_{\|x-x'\| < \epsilon} \mathcal{L}(f(x'), y) \quad (1)$$

where $\mathcal{L}$ is the loss function and ϵ is the perturbation budget that constrains the strength of the attack under an ℓ_p -norm.

Adversarial Training AT is a defense method that improves model robustness by training on adversarially perturbed samples alongside clean samples. The objective for AT is usually formulated by:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\delta: \|\delta\|_p \leq \epsilon} \mathcal{L}(f(x'; \theta), y) \right] \quad (2)$$

where $\mathcal{D}$ is the data distribution.

Adversarial Pruning Given a pre-trained model $f(\theta)$ under AT, we aim to maintain robustness while applying a pruning mask M across layers to achieve compression. The general pruning is carried out by removing parameters until the network reaches a desired s_r . So, pruning involves creating M , formalized as follows:

$$M = \arg \min_{\|m\|_0 \leq 1-s_r} \mathcal{L}(f(\theta) \otimes m, x, y). \quad (3)$$

$$m = \begin{cases} 0, & \text{if } \theta_i \text{ is to be pruned} \\ 1, & \text{otherwise} \end{cases}, \quad \forall \theta_i \in \theta \quad (4)$$

For AP, we follow the strategy from HARP[52]. To enhance adversarial robustness, inner maximization generates adversarial examples, targeting a worst-case loss under the PGD attacks:

$$\max_{\delta} \mathcal{L}_r(f(\theta) \odot M, x + \delta, y) \quad (5)$$

where $\mathcal{L}_r$ defines the robust loss function, incorporating attack-specific parameters. And the global non-uniform pruning strategy jointly optimizes a compression rate c_r and importance scores I across layers. This is defined by:

$$\min_{c_r, I} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\delta} \mathcal{L}_r(f(\theta) \odot M, x + \delta, y) \right] + \varphi \cdot \mathcal{L}_p(f(\theta) \odot M, a_t) \quad (6)$$

where $\mathcal{L}_p$ enforces target compression a_t across all layers L to guide non-uniform pruning rates.

3.2. Pruning Importance Score

The importance score's definition varies by pruning method. This section details its definition and computation, as discussed in Sec.3.1.

Neuron Importance Score (NIS) NIS [50] is a common pruning score that minimizes the weighted distance between the original and pruned final responses in a given layer. Given a neural network $f^{(n)}$ with n layers, and the importance score of the final layer response $I^{(n)}$, the importance score of the i -th layer $I^{(i)}$ is calculated as follows:

$$I^{(i)} = |w^{(i+1)}|^T \cdot |w^{(i+2)}|^T \dots |w^{(n)}|^T \cdot I^{(n)} \quad (7)$$

Here, $w^{(i)}$ refers to the weight matrix of the i -th layer. In this case, the importance of a neuron in the network can be computed recursively along the network. Thus, $I^{(i)}$ can be propagated from the importance score of the $(i+1)$ -th layer as follow:

$$I_{NIS}^{(i)} = |w^{(i+1)}|^T \cdot I^{(i+1)} \quad (8)$$

Empirical Risk Minimization (ERM) The importance score based on ERM, used in HYDRA[34] and HARP, is defined as follows, based on the weight magnitudes of a pre-trained model.

$$I_{ERM}^{(i)} = \frac{\eta \cdot |\theta^{(i)}|}{\max(|\theta^{(i)}|)} \quad (9)$$

Here, $I_{ERM}^{(i)}$ represents the importance score for the weights of the i -th layer, $\theta^{(i)}$ denotes the weight matrix of that layer, and η is a scaling factor determined by the layer's structure and receptive field size.

Adversarial Saliency Estimation (ASE) MAD [21] computes the ASE for each parameter, removing the least salient ones. Adversarial saliency is estimated via a second-order Taylor expansion, with the saliency score I_{ASE} for each parameter $w^{(i)}$ defined as:

$$I_{ASE} = \frac{1}{2} \left[\frac{\partial^2 \mathcal{L}}{\partial (w^{(i)})^2} \right] |w^{(i)}|^2 \quad (10)$$

Here, $\frac{\partial^2 \mathcal{L}}{\partial (w^{(i)})^2}$ is the Hessian of the loss function with respect to $w^{(i)}$, reflecting the impact of parameter changes on the loss, and $|w^{(i)}|^2$ reflects the magnitude and importance of the corresponding parameter.

Batch Normalization Scaling Factor (BNSF) BNAP [43] identifies the magnitude of the Batch Normalization (BN) scaling factor γ as a robustness measure across channels, used as the pruning criterion. The BN layer normalizes weights w_{ik}^j into the following effective weights:

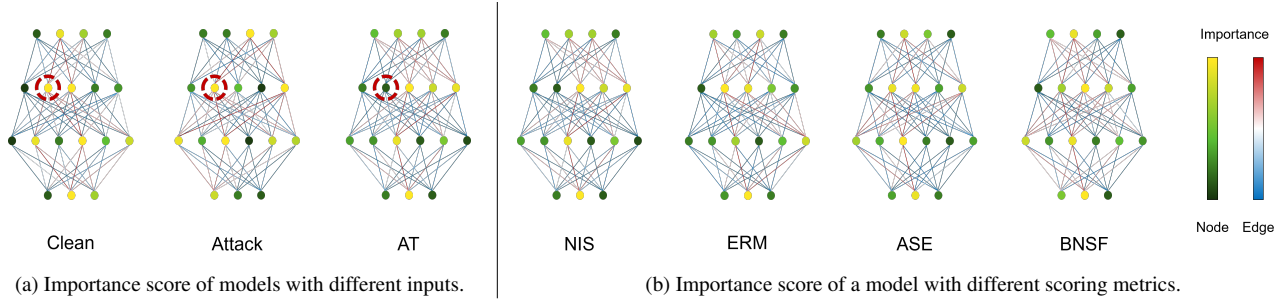


Figure 2. Importance score visualization for the 4-5-6-3 CNN model on the MNIST dataset: yellow/red indicate higher importance, green/blue indicate lower.

$$\hat{w}_{ik}^j = \frac{\gamma_i^j w_{ik}^j}{(\sigma_i^j)^2 + \epsilon} \approx \frac{\gamma_i^j w_{ik}^j}{\sigma_i^j} \quad (11)$$

where i is the layer index, j is the channel index of i -th layer, and k is the channel index in the $(i - 1)$ -th layer. In this case, ϵ is small enough to be neglected, and γ_i^j and σ_i^j each play a role in readjusting the weights per channel. Therefore, the computed weights can be used as the importance score.

3.3. A Closer Look at Adversarial Pruning via Importance Score

Least Importance Score [50] is a common pruning criterion that assesses each parameter’s (weights, filters, channels) importance to overall network performance, pruning the least important first. This score clarifies the impact of adversarial examples on pruning. We compared importance scores between models trained on clean data, adversarial data, and adversarially-trained models using a simple CNN architecture.

As shown in Fig. 2a, importance scores differ significantly between models trained on clean versus adversarial datasets. In the AT model, trained on both, scores do not consistently fall between clean and adversarial values but more closely resemble the clean model. Notably, nodes highly important in both cases (e.g., the second node in the second layer) may appear less important in the AT model. This suggests that training on both datasets may obscure importance, potentially leading to pruning of critical nodes or connections.

Moreover, we evaluated the same model with different importance scoring methods mentioned earlier. The results in Fig. 2b suggest that the order of importance for nodes and connections can vary depending on the scoring method. In other words, each sub-network pruned with a different importance scoring method may contain important information that is not shared by the others. This implies that sub-models sharing the same base model could act as multiple models in ED, while complementing each other.

4. Efficient Ensemble Defense (EED)

A common hypothesis in adversarial ensemble defense is that training and combining multiple base models enhances adversarial robustness compared to a single model [7]. However, traditional ED methods often increase model capacity, which is counterproductive for reducing computational and memory costs. But is it truly impossible to achieve an efficient adversarial ensemble defense for a compact model?

We propose a novel Efficient Ensemble Defense (EED) method that leverages model compression while maintaining robustness. Sec. 3.3 has demonstrated the potential for sub-models, pruned based on different importance scores from a single base model, to act as multiple base models in an ensemble. By ensembling these sub-models, we can reduce model capacity compared to traditional ED. However, it is observed that the smaller the model during ensemble, the more the robustness performance decreases; thus, high diversity is required to mitigate this issue[30]. Therefore, we enhance the ensemble effect and maintain robustness through robust diversity evaluation and misclassification regularization. Furthermore, to improve inference speed—a primary goal of model compression—we dynamically add sub-models during inference and perform ensemble only as needed, thus enhancing inference efficiency.

EED is designed with three main considerations: 1) the diversity of pruned sub-models, 2) improvement of robustness during the ensemble process, and 3) maintaining compactness throughout the ensemble process.

4.1. Enhancing Robust Diversity

Our strategy begins with generating sub-models from a base model through pruning, where each sub-model undergoes an AP process based on a unique importance score. This AP adheres to the principles described in Sec. 3.1, utilizing four importance scoring methods detailed in Sec. 3.2. To further enhance diversity, we divide the training dataset for pruning into multiple subsets, with each subset designated to train a specific sub-model. Some subsets are shared across all

Algorithm 1 Efficient Ensemble Defense (EED)

Input: Training data D , base model M
Output: Ensemble $Selected$
for each subset $D_i \subset D$ **do**
 $F_i \leftarrow \text{AdversarialPruning}(M, D_i)$
 $Set(D) \leftarrow Set(D) \cup F_i$
end for
 $EnsSet \leftarrow \text{FormEnsembleSet}(Set(D))$
for each ensemble $E \in EnsSet$ **do**
 $RD \leftarrow \text{CalculateRobustDiversity}(E)$
end for
 $Selected \leftarrow \text{SelectEnsemble}(EnsSet, RD)$
for each training example $(x, y) \in D$ **do**
 $\mathcal{L}_{EED}(x, y) \leftarrow \mathcal{L}_E + \mathcal{L}_R + \mathcal{L}_C$
 $UpdateModels(Selected, \mathcal{L}_{EED})$
end for
 {————— **INFERENCE PHASE** —————}
for each test example $x \in T$ **do**
 $pred \leftarrow \text{InitializePredictions}(Selected)$
 $t \leftarrow 1$
 while not sufficient robustness **do**
 $pred \leftarrow \text{UpdateEnsemble}(pred, Selected[t])$
 $t \leftarrow t + 1$
 $q_t \leftarrow \text{CalculateStoppingCriterion}(pred)$
 if robustness criterion met **then**
 break
 end if
 end while
end for
Output $pred$

sub-models to retain common core knowledge, while the remaining subsets are distributed across different sub-models to ensure each to learn somewhat different aspects of the data. These subsets are randomly reshuffled whenever the importance scoring method changes. This approach amplifies model diversity, as each sub-model becomes specialized in defending against attacks targeting specific patterns or features in the data. Ultimately, we establish a model pool composed of sub-models with varying importance scoring methods or trained on different datasets.

Let this be denoted as pool of N models trained on the dataset D , represented as $Set(D) = \{F_0, \dots, F_{N-1}\}$. The ensemble set $EnsSet$ represents the collection of all possible ensemble teams with sizes S ranging from 2 to N formed from $Set(D)$. The number of ensemble teams is given by $|EnsSet| = \sum_{S=2}^N \binom{N}{S} = 2^N - (1 + N)$.

To quantify the diversity among sub-models within each ensemble team, we define a Robust Diversity (RD) score based on the probability of defense failure for the model. The diversity score relies on two probabilities: p_{one} and p_{two} . p_{one} is the probability that a randomly chosen model

fails to defend (viz. the attack succeeds), while p_{two} is the expected probability that two randomly selected models both fail to defend:

$$p_{one} = \sum_{i=1}^S \frac{S_i}{S} p_i, \quad p_{two} = \sum_{i=1}^S \frac{S_i(i-1)}{S(S-1)} p_i \quad (12)$$

Here, p_i represents the defense failure probability of the i -th model among the total S models. Based on the two probabilities, the RD score is defined by:

$$RD = 1 - \frac{p_{two}}{p_{one}} \quad (13)$$

This RD metric measures diversity. $RD = 1$ indicates maximum diversity (i.e., even if one model fails in defense, the likelihood of other models succeeding is high), while $RD = 0$ indicates no diversity (i.e., both models fail in defense). The EED framework applies RD-based ensemble selection to form the smallest ensemble from the $EnsSet$ of sub-models with the high RD scores.

4.2. Regularization against Misclassification

As in the existing ED technique [47], the classifier is trained by combining the classification loss, regularization(Reg) and diversity(Div) terms. Specifically, for a single training example (x, y) , the ensemble loss function is defined by:

$$\mathcal{L}_E(x, y) = \sum_{i=1}^N \ell(F_i(x), y) + \alpha Reg(h(x)) + \beta Div(h^1(x), h^2(x), \dots, h^N(x), y) \quad (14)$$

Here, α and β are hyperparameters, and $h^i(x) = (h_1^i(x), \dots, h_C^i(x))$, $i \in \mathbb{N}$ represents the output of the i -th classifier. The final output is calculated using the average combiner $h(x) = \frac{1}{N} \sum_{i=1}^N h^i(x)$.

To improve the robustness of EED, we employ an additional regularization term that focuses on the probability scores of the most misclassified class, inspired by iGAT [7]. For a given example (x, y) , the proposed term is defined by:

$$\mathcal{L}_R(x, y) = -\psi_{0/1}(c(h^1(x), \dots, h^N(x)), y) \cdot \log \left(1 - C \max_{i=1}^N \max_{j=1}^C h_i^j(x) \right) \quad (15)$$

The error function $\psi_{0/1}(f, y) = \frac{1}{|D|} \sum_{x \in D} 1[f_y(x) < \max_{c=y} f_c(x)]$ returns 0 if f correctly predicts the label y , and 1 otherwise. This regularization term plays a crucial role in guiding the model to reduce misclassifications, especially improving its ability to handle ambiguous data points or difficult input examples. By doing so, each base classifier focuses more on the examples it misclassified, enhancing the robustness of the overall ensemble model across diverse input spaces.

4.3. Regularization for Compactness

Another key objective of the EED framework is maintaining the compressed models' compactness while maximizing the ensemble merit. To achieve this, we design a compactness regularization term $\mathcal{L}_C$ to maintain weight and activation sparsity among the sub-models, thereby improving the compactness of the ensemble. The weight regularization term controls the sparsity of the sub-model weights, ensuring that they remain sparse above a certain threshold, thus suppressing model capacity growth. Specifically, the weight sparsity regularization term $\mathcal{L}_W$ is defined as follows:

$$\mathcal{L}_W(x, y) = \sum_{i=1}^N \|W_i\|_1, \quad (16)$$

Here, $\|W_i\|_1$ denotes the L_1 -norm of the weights W_i of the i -th sub-model, which encourages sparsity by minimizing the sum of the absolute values of the weights.

To control the activation sparsity of the sub-models, we use an additional sparsity regularization term $\mathcal{L}_A$, which encourages each sub-model to reduce unnecessary neuron activations. This term is defined as follows:

$$\mathcal{L}_A(x, y) = \sum_{i=1}^N \|h^i(x)\|_0, \quad (17)$$

Here, $\|h^i(x)\|_0$ represents the sum of the number of non-zero activations in the output $h^i(x)$ of the i -th sub-model.

Thus, $\mathcal{L}_C$ can be summarized as follows:

$$\mathcal{L}_C(x, y) = \lambda_1 \mathcal{L}_W + \lambda_2 \mathcal{L}_A \quad (18)$$

The hyperparameter λ_1 regulates weight sparsity, with higher values leading to more sparse weights and reduced model capacity. Similarly, λ_2 controls activation sparsity, making neuron activations sparser, which results in a more compact model structure that saves memory and computational resources.

Finally, the overall loss function of EED is as follows:

$$\mathcal{L}_{EED}(x, y) = \mathcal{L}_E(x, y) + \omega \mathcal{L}_R(x, y) + \gamma \mathcal{L}_C(x, y) \quad (19)$$

The EED framework provides high robustness against adversarial attacks while ensuring compactness through these multiple regularization terms, making it suitable to efficient applications like real-time inference.

4.4. Dynamic Ensemble for Inference Efficiency

In the training phase, EED optimizes all sub-models to ensure diversity and robustness. To enhance the compactness and efficiency of EED during the inference phase, we adopt an inference-efficient ensemble method [25]. We apply this to adversarial defense by dynamically constructing an ensemble of varying sizes for each sample, termed Dynamic

Inference Ensemble (DIE). Specifically, for a given sample x , EED dynamically determines the optimal point to stop while adding each sub-models progressively. As a result, by monitoring the uncertainty of each sub-model on adversarial examples and only introducing additional models when necessary, we avoid unnecessary computations while maintaining high robustness.

DIE sequentially adds sub-models and evaluates the robustness of the predictions at each stage. Starting with the first sub-model F_1 , a new sub-model F_t is added at each stage t , and the intermediate ensemble prediction $\hat{y}_{ens}^t$ is updated. These intermediate predictions serve as a criterion for assessing how robust the added sub-models are. The intermediate ensemble prediction based on the predictions $\hat{y}_i$ of each sub-model F_i is computed by:

$$\hat{y}_{ens}^t = \frac{1}{t} \sum_{i=1}^t \hat{y}_i \quad (20)$$

Here, each $\hat{y}_{ens}^t$ represents the combined prediction of the sub-models for the input (both clean and adversarial examples), and it plays a role in strengthening the overall robustness of the ensemble.

At each stage t , the early stopping probability q_t is assessed. If the current ensemble performance is determined to be robust enough, the addition of sub-models is halted. The value of q_t is computed based on the prediction uncertainty and confidence. The prediction uncertainty is computed by KL-divergence of t and $(t-1)$ -th ensemble predictions, and the prediction confidence setting follows WoC[40]. The optimal stopping point z_x is defined as follows, based on a geometric distribution modeling with discrete Bernoulli trials:

$$z_x = \arg \max_t \left(q_t \prod_{i=1}^{t-1} (1 - q_i) \right) \quad (21)$$

Here, q_t is dynamically adjusted at each stage based on robustness, and sub-models are added only when necessary, allowing for efficient use of computational resources.

The final ensemble prediction $\hat{y}_{ens}$ is calculated as the average of the predictions of all sub-models up to the stopping point z_x :

$$\hat{y}_{ens} = \frac{1}{z_x} \sum_{i=1}^{z_x} \hat{y}_i \quad (22)$$

This approach takes into account the predictions of all sub-models while avoiding the inclusion of unnecessary ones, thereby enhancing efficiency. By applying different ensemble sizes for each sample, the overall computational cost is reduced while still being robust against adversarial attacks.

DIE finds a balance between efficiency and robustness, quickly identifying the optimal combination of sub-models to ensure strong performance even in real-time inference

Table 1. Comparison of various AP methods and EED on CIFAR-10 and SVHN datasets when sparsity rate $s_r = 80\%$

	Setting	CIFAR-10						SVHN						Prms
		Clean	PGD	AA	C&W	DF	Speed up	Clean	PGD	AA	C&W	DF	Speed up	
Resnet 18	AT[27]	87.05	56.14	48.02	57.60	53.10	1.00x	93.37	56.27	50.14	58.85	59.80	1.00x	11.2M
	R-ADMM[48]	81.25	48.00	43.92	49.17	39.11	1.68x	74.81	49.73	37.40	52.62	43.40	1.63x	2.2M
	HYDRA[34]	77.36	<u>52.92</u>	43.74	49.64	45.91	1.77x	91.06	52.22	<u>47.62</u>	55.13	46.13	1.74x	
	RST[10]	61.02	41.01	18.38	51.02	26.82	1.86x	82.39	46.29	36.35	52.65	39.27	<u>1.83x</u>	
	MAD[21]	74.18	50.38	41.27	54.17	38.44	1.69x	92.84	51.65	39.87	59.80	44.72	1.70x	
	Flying Bird[2]	81.07	51.62	44.41	56.08	45.29	1.79x	90.21	52.06	42.01	57.30	<u>52.23</u>	1.76x	
	HARP[52]	83.84	52.56	<u>45.36</u>	56.57	47.04	1.81x	92.60	<u>54.16</u>	45.89	57.28	51.24	1.80x	
	TwinRep[23]	77.26	52.04	<u>43.52</u>	<u>56.60</u>	46.18	<u>1.82x</u>	92.96	53.83	44.73	56.26	48.61	1.78x	
	EED(ours)	86.13	55.71	48.13	57.03	51.97	1.86x	93.15	55.74	50.18	<u>58.37</u>	56.05	1.85x	
VGG 16	AT[27]	82.70	54.49	48.52	54.91	56.91	1.00x	93.06	57.64	52.28	55.36	61.49	1.00x	14.7M
	R-ADMM[48]	76.51	47.19	40.41	44.80	48.29	1.65x	65.40	48.39	44.64	48.92	49.85	1.66x	2.9M
	HYDRA[34]	78.83	47.85	44.85	45.73	49.41	1.72x	90.82	53.66	46.30	52.63	53.42	1.72x	
	RST[10]	77.29	39.90	30.15	32.02	42.03	1.83x	82.39	41.94	45.52	43.58	38.81	1.81x	
	MAD[21]	73.65	48.81	40.88	47.16	45.50	<u>1.80x</u>	89.94	49.46	40.31	53.03	47.37	1.69x	
	Flying Bird[2]	77.36	49.43	<u>45.73</u>	50.38	48.34	1.68x	90.11	53.29	49.18	52.81	50.78	1.73x	
	HARP[52]	<u>80.92</u>	<u>51.77</u>	44.02	<u>52.79</u>	<u>52.17</u>	1.73x	93.22	<u>55.53</u>	44.73	<u>52.96</u>	56.16	<u>1.77x</u>	
	TwinRep[23]	76.04	50.12	43.26	52.42	51.59	1.79x	92.60	54.61	49.06	51.24	56.58	1.75x	
	EED(ours)	81.39	54.26	47.49	53.27	53.94	1.79x	<u>93.14</u>	57.19	51.49	53.12	60.27	<u>1.77x</u>	

settings. As a result, the EED framework can lower the average inference cost in adversarial attack scenarios while maintaining high level of robustness, making it an effective defense, even in resource-constrained environments.

5. Experimental Evaluations

5.1. Experiment Setups

Classifiers and Datasets We utilize two specific architectures, ResNet18 and VGG16, as classifiers and conduct experiments focusing on CIFAR-10 and SVHN. We divide the dataset into d subsets for pruning into various sub-models, setting $d = 4$ to create three sub-models for each pruning metric. 25% of the dataset is shared across all sub-models, while the remaining 75% is allocated separately for each of the three sub-models. With four pruning metrics in play, we end up with a total of $N = 12$ sub-models, which together form the ensemble set, denoted as *EnsSet*. For ensemble learning, the entire dataset is used.

Attacks We use PGD-20 attacker with a step size of $2/255$ to generate adversarial examples for training, and Auto Attack(AA)[6], C&W[1] and Deepfool(DF)[28] for evaluation. Except for DF using l_2 norm, the rest use l_∞ norm, and all attacks use perturbation strength $\epsilon = 8/255$.

Training We use the Madry[27] as AT method. The batch size is set to 512, and the multi-step learning rates are $\{0.01, 0.002\}$ for CIFAR-10 and $\{0.1, 0.02, 0.004\}$ for SVHN. The hyperparameters are set as follows: $\omega = 10$, $\lambda_1 = 0.7$, $\lambda_2 = 0.25$, and the RD threshold is set to 0.7. c_r is set to be more than four times sparser than the desirable s_r , considering the size of N , and γ is adjusted to ensure that the s_r is met even after the ensemble. (When $s_r=80\%$, we set $c_r = 95\%$ and $\gamma = 4$.) Each experiment is run on an NVIDIA RTX A6000 GPU with 8 CPU cores.

5.2. Result Comparison and Analysis

This section compares the proposed EED with existing APs for sparsity rate(s_r) of 80% on the CIFAR-10 and SVHN datasets. Table 1 demonstrates that EED consistently provided superior adversarial robustness, with significantly higher accuracy against attacks such as PGD, AA, C&W, and DF compared to other methods. Additionally, EED maintained high performance on clean data, indicating a proper balance between adversarial robustness and clean accuracy.

Note that EED demonstrates superior robustness, particularly against strong attacks such as AA and PGD, outperforming existing defense methods like HYDRA, HARP, and TwinRep. For example, in the CIFAR-10 experiment using the ResNet-18 architecture, EED achieved 55.71% accuracy against PGD and 48.13% against AA, showing an improvement of over 2% compared to other methods and achieving the highest robustness. Moreover, despite using only 20% of the model's parameters, the performance drop in AT was minimal, less than 0.5%. Also, EED showed stronger defense capabilities than AT against PGD in the ResNet-18-based SVHN experiment.

As a result, EED accomplished remarkable performance improvements without sacrificing efficiency. Notably, it also records up to 1.86 times faster inference speeds, comparable to or exceeding existing techniques. This is crucial for practical applications where computational resources may be limited, demonstrating that EED is an ideal defense method for resource-constrained environments. Furthermore, EED showed consistent performance across different model architectures such as ResNet-18 and VGG-16, suggesting that EED can be applied to a wide range of network architectures with minimal adjustment.

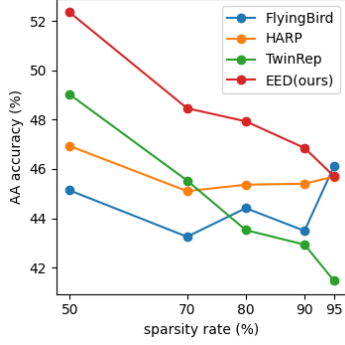


Figure 3. Robust performance by s_r .

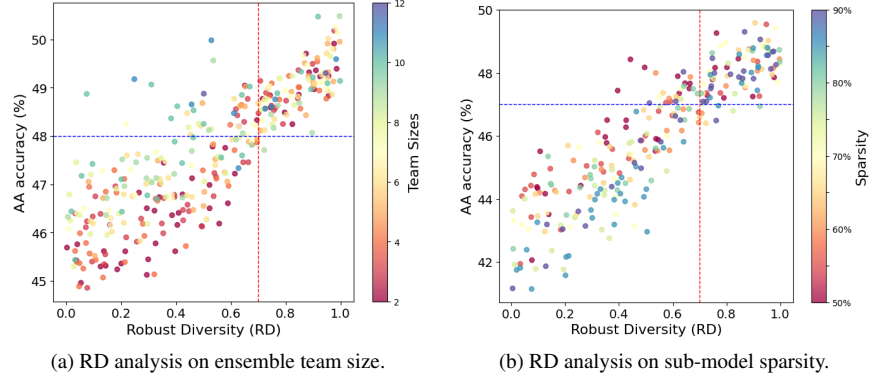


Figure 4. Correlation between RD and robustness in different ensemble components.

5.3. Ablation Studies

Ablation studies are conducted on Resnet18 and CIFAR-10. **Changes along Sparsity** Fig.3 shows the change in robustness performance with respect to the sparsity rate. EED demonstrated the highest performance compared to the existing top three methods, except for a very high s_r (95%) against AA. EED tended to show a decrease in performance as s_r increases, which is believed to be due to the reduction in ensemble diversity caused by the higher compression rate. On the other hand, at lower s_r , EED exhibited higher robustness than AT due to the enhanced ensemble diversity. More detailed experimental results for different s_r can be found in the supplementary materials.

Analyzing Robust Diversity Fig.4 analyzes the role of RD in EED. Fig.3a, which shows robustness according to ensemble team size, demonstrates that a high RD is necessary for small ensemble sizes to achieve high performance. Fig.3b shows robustness from the RD perspective with respect to the compression rate (c_r) of the sub-models. We can find that to maintain high performance while keeping sub-models small, a high RD is also required.

Analyzing Regularization Components EED has two regularization terms, as mentioned in Sec.4.2 and Sec.4.3. Regularization against misclassification (L_R) is controlled by ω , while regularization for compactness (L_C) is controlled by γ . This section analyzes the impact of each term on the performance of EED. Tab.2 shows that L_R has a greater effect on clean performance, while L_C has a larger impact on attack performance. In particular, when the compactness term is not used, attack performance significantly deteriorates, which is likely due to the need of reducing the ensemble size further to meet the s_r , thereby diminishing the ensemble advantages.

Effect of Dynamic Inference Ensemble Tab.3 analyzes the effect of DIE for different sparsity rates. At $s_r = 80\%$, DIE had little impact on the main performance and only showed a slight speed improvement. On the other hand, at $s_r = 50\%$, activating DIE resulted in performance im-

Table 2. Analysis of EED with regularization terms.

	Settings	Clean	PGD	AA
CIFAR-10	EED	86.13	55.71	48.13
	$\omega = 0$	74.16	47.45	36.37
Resnet 18	$\gamma = 0$	76.33	39.10	31.59
	$\omega, \gamma = 0$	73.95	37.83	28.94

Table 3. Correlation between dynamic inference ensemble and s_r .

	Settings	Clean	PGD	AA	Speed up
$s_r = 80\%$	w/ dynamic	86.13	55.71	48.13	1.86x
	w/o dynamic	86.20	55.62	48.11	1.80x
$s_r = 50\%$	w/ dynamic	86.15	56.09	52.35	1.64x
	w/o dynamic	85.93	54.28	50.76	1.27x

provements of 1.81% and 1.59% for PGD and AA, respectively, while significantly improving inference speed. This suggests that DIE is more effective at lower sparsity. The reason for this performance improvement is that, with lower sparsity, the ensemble can be larger, thus enhancing the optimization effect of the dynamic ensemble. In contrast, without DIE, the likelihood of conflicts between the sub-models may increase as the ensemble grows larger.

6. Conclusion

This paper proposes the Efficient Ensemble Defense (EED) technique to maintain both compression and robustness of deep learning models. EED constructs an ensemble of sub-models using different pruning importance scores to enhance adversarial robustness while reducing model size and computational cost. By addressing information loss and limited model diversity caused by pruning with ensemble methods and regularization, EED outperforms existing AP methods in accuracy and robustness. Experimental results show superior resilience of EED against various attacks on the CIFAR-10 and SVHN, maintaining inference efficiency. EED also proves its applicability in diverse neural network architectures and resource-constrained environments.

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